

QUESTION 1

27 MARKS

Simplify the following fully, without the use of a calculator, and showing all working. Leave your answers in surd form and with a rational denominator where necessary.

a) $81^{\frac{5}{4}}$ (2)

$$= (3^4)^{\frac{5}{4}}$$

$$= 3^5$$

$$= 243 \checkmark$$

b) $(4 - 2\sqrt{5})(5 + 3\sqrt{5})$ (2)

$$= 20 + 12\sqrt{5} - 10\sqrt{5} - 6(5)$$

$$= -10 + 2\sqrt{5} \checkmark$$

c) $\frac{11^{2025} - 11^{2023}}{11^{2022} - 11^{2021}}$ (4)

$$= \frac{11^{2023}(11^2 - 1)}{11^{2021}(11 - 1)} \checkmark$$

$$= \frac{11^2(121 - 1)}{10}$$

$$= \frac{121(120)}{10} \checkmark$$

$$= (121)(12)$$

$$= 1452 \checkmark$$

$$d) \frac{54^{x+1} \cdot 36^{x-1}}{242^{x-3}} \times \left(\frac{2}{3}\right)^{3x} \quad (4)$$

$$= \frac{(3^3 \times 2)^{x+1} \cdot (3^2 \times 2^2)^{x-1} \cdot 2^{3x}}{(3 \times 2^3)^{2x-3} \cdot 3^{3x}}$$

$$= \frac{3^{3x+3} \cdot 2^{x+1} \cdot 3^{2x-2} \cdot 2^{2x-2} \cdot 2^{3x}}{3^{2x-3} \cdot 2^{6x-9} \cdot 3^{3x}} \quad \checkmark$$

$$= 2^{x+1+2x-2+3x-(6x-9)} \cdot 3^{3x+3+2x-2-(2x-3)-3x} \quad \checkmark$$

$$= 2^8 \cdot 3^4 \quad \checkmark$$

$$= 20736$$

$$e) \frac{5 \cdot 3^x - 9 \cdot 3^{x-2}}{3^x - 3^{x-1}} \quad (4)$$

$$= \frac{5 \cdot 3^x - 3^2 \cdot 3^x \cdot 3^{-2}}{3^x - 3^x \cdot 3^{-1}} \quad \checkmark$$

$$= \frac{3^x (5 - 1)}{3^x (1 - 3^{-1})} \quad \checkmark$$

$$= \frac{4}{1 - \frac{1}{3}} \quad \checkmark$$

$$= \frac{4}{\frac{2}{3}}$$

$$= 6 \quad \checkmark$$

$$f) \frac{\sqrt{2n} \times \sqrt[6]{2n+2}}{\sqrt[3]{4n-4}} \quad (3)$$

$$= \frac{2^{\frac{n}{2}} \cdot 2^{\frac{n+2}{6}}}{(2^2)^{\frac{n-4}{3}}} \quad \checkmark$$

$$= 2^{\frac{n}{2} + \frac{n+2}{6} - \frac{(2n-8)}{3}} \quad \checkmark$$

$$= 2^{\frac{3n + n + 2 - 4n + 16}{6}}$$

$$= 2^{\frac{18}{6}}$$

$$= 2^3$$

$$= 8 \quad \checkmark$$

$$g) \frac{2^{n+1}}{(2^n)^{n-1}} \div \frac{4^{n+1}}{(2^{n-1})^{n+1}} \quad (5)$$

$$= \frac{2^{n+1}}{2^{n^2-n}} \times \frac{2^{n^2-1}}{2^{2n+2}} \quad \checkmark$$

$$= 2^{n+1+n^2-1-(n^2-n)-(2n+2)}$$

$$= 2^{-2} \quad \checkmark$$

$$= \frac{1}{4}$$

$$h) \left(\frac{1}{x} + \frac{x}{y}\right)^{-1}$$

(3)

$$= \left(\frac{y + x^2}{xy}\right)^{-1}$$

$$= \frac{xy}{y + x^2} \quad \checkmark \checkmark$$

QUESTION 2

13 MARKS

Solve for x :

a) $2^x = \sqrt{0,125}$ (2)

$$2^x = \left(\frac{1}{8}\right)^{\frac{1}{2}}$$

$$2^x = 2^{-\frac{3}{2}} \quad \checkmark$$

$$\therefore x = -\frac{3}{2} \quad \checkmark$$

b) $5^{x+2} + 5^x = 26$ (3)

$$5^x(5^2 + 1) = 26 \quad \checkmark$$

$$5^x(26) = 26$$

$$5^x = 1 \quad \checkmark$$

$$\therefore x = 0 \quad \checkmark$$

c) $2 \cdot 2^{2x} - 9 \cdot 2^x + 4 = 0$ (4)

$$\text{let } k = 2^x$$

$$2k^2 - 9k + 4 = 0$$

$$(2k - 1)(k - 4) = 0 \quad \checkmark$$

$$k = \frac{1}{2}$$

$$\text{OR } k = 4$$

$$2^x = \frac{1}{2}$$

$$2^x = 4$$

$$\therefore x = -1 \quad \checkmark$$

$$\therefore x = 2 \quad \checkmark$$

d) $x^{\frac{1}{2}} - x^{\frac{1}{4}} - 2 = 0$

(4)

Let $k = x^{\frac{1}{4}}$

$k^2 - k - 2 = 0$ ✓

$(k-2)(k+1) = 0$ ✓

$k = 2$ OR $k = -1$

$x^{\frac{1}{4}} = 2$

$x^{\frac{1}{4}} = -1$ ✓

$x = 2^4$

No Soln.

$x = 16$ ✓

QUESTION 3

5 MARKS

Solve for x and y :

$8^x = 4^{2+y}$

$9 \cdot 3^y = 27^x$

(5)

$2^{3x} = 2^{2(2+y)}$

$\therefore 3x = 4 + 2y$ ✓ (A)

$3^2 \cdot 3^y = 3^{3x}$

$3^{y+2} = 3^{3x}$

$\therefore y+2 = 3x$ ✓ (B)

Let (A) = (B) ✓ M

$y+2 = 4+2y$

$-2 = y$ ✓

$\therefore x = 0$ ✓